Progressive Education Society'sModern College of Engineering, Pune-05.(An Autonomous Institute Affiliated to Savitribai Phule Pune University) Department of S.Y. B. Tech [AIDS]

Assignment No: 08

SUBJECT: Java Programming

Name: Om Mane

Class: SY BTech (AIDS)

Section: A

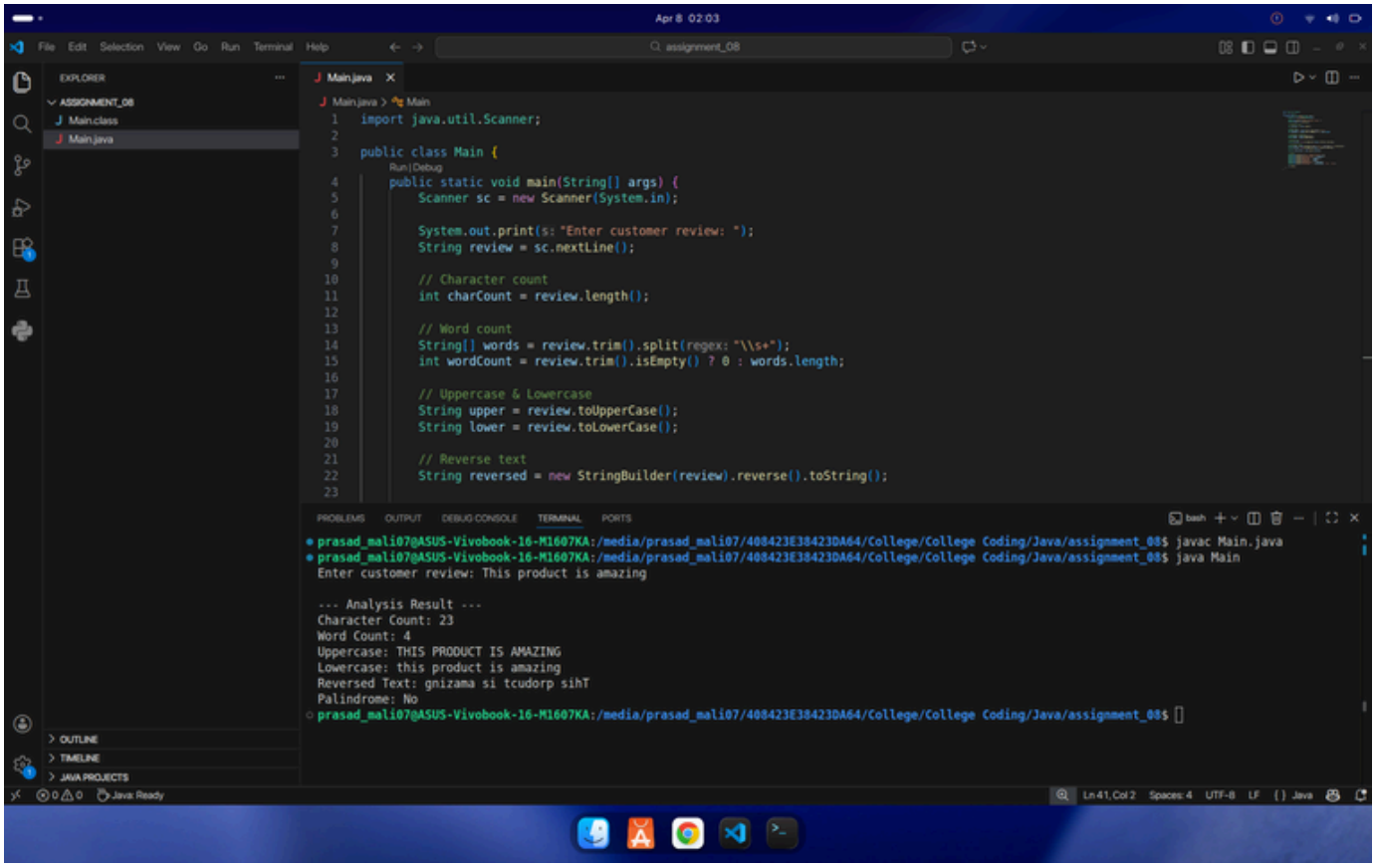
Roll Number: 26040

Title / Problem Statement :

Develop an application that empowers e-commerce businesses to gain deeper insights from customer reviews by extracting key textual features: word and character counts, normalized text (uppercase/lowercase), reversed text for inspection, and a palindrome flag to highlight unique review structures.

Input :

```java
import java.util.Scanner;


public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.print("Enter customer review: ");

        String review = sc.nextLine();


        // Character count

        int charCount = review.length();


        // Word count

        String[] words = review.trim().split("\\s+");

        int wordCount = review.trim().isEmpty() ? 0 : words.length;


        // Uppercase & Lowercase

        String upper = review.toUpperCase();

        String lower = review.toLowerCase();

        // Reverse text

        String reversed = new StringBuilder(review).reverse().toString();

        // Palindrome check

        String clean = review.replaceAll("\\s+", "").toLowerCase();

        String revClean = new StringBuilder(clean).reverse().toString();

        boolean isPalindrome = clean.equals(revClean);

        // Output

        System.out.println("\n--- Analysis Result ---");

        System.out.println("Character Count: " + charCount);
```

```java
        System.out.println("Word Count: " + wordCount);

        System.out.println("Uppercase: " + upper);

        System.out.println("Lowercase: " + lower);

        System.out.println("Reversed Text: " + reversed);

        System.out.println("Palindrome: " + (isPalindrome ? "Yes" : "No"));


        sc.close();

    }
```

Output :